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INTRODUCTION

s out different functions that can be grouped into the following categories. Metabolic: glycogenesis, glycogenolysis, gluconeogenesis, transamination, deamination, ureogenesis, lipogenesis, β -oxidation of fatty acids, ketogenesis, and synthesis of lipoproteins and phospholipids (Wang *et al.*, 2019; Habel and Sundrum, 2020; Lopreiato *et al.*, 2020; McFadden, 2020; Zhang *et al.*, 2020; Huang *et al.*, 2022). Vascular: blood storage, and lymph synthesis (Gross and Bruckmaier, 2019a; Mezzetti *et al.*, 2020). Secretory bilirubin, bile acids and electrolytes (Gross and Bruckmaier, 2019b; Allen, 2020; Zachut *et al.*, 2020). Excretory: steroid hormones, calcium ion (Ca^{2+}), antibiotics, and drug metabolites (Macrae *et al.*, 2019; Paiano *et al.*, 2019; Haussler *et al.*, 2022). The analysis of glucose (**GLU**), urea, aspartate aminotransferase (**AST**), albumin (**ALB**), alanine aminotransferase (**ALT**), total bilirubin (**TB**), conjugated bilirubin (**CB**), cholesterol (**CHOL**) and γ -glutamyl-transferase (**γ -GT**), forms a general assessment of liver blood markers (Paiano *et al.*, 2020) and provides information on cow productivity (Pascottini *et al.*, 2022). Also, milk production and quality results from internal metabolic balance (McFadden, 2020). Therefore, they can also quantify different milk parameters such as milk fat, solids-non-fat (**SNF**), milk density, milk protein, cryoscopic point (**CRP**), pH, and somatic cell count (**SCC**) (Stocco *et al.*, 2020; France *et al.*, 2022). Clinical and laboratory monitoring in dairy cows generally involves the 1st and 2nd third of lactation, especially for the detection of subclinical forms of ketosis and ruminal acidosis (García *et al.*, 2020; Pinedo and Melendez, 2022). Because the risk of such pathologies during the 3rd third of lactation is very low, blood samples are rarely monitored at this time (Gross and Bruckmaier, 2019a; Yang *et al.*, 2019). Therefore, the present study aims to evaluate the relationships between liver blood markers and milk parameters in cows during the 1st and 2nd third of lactation.

MATERIALS AND METHODS

In multiparous cows, the period of peak milk production is usually between 30 to 60 d p.p., when the uterus is involuting and the ovary is returning to estrous cyclicity (Davidson and Stabenfeldt, 2014; Fails and Magee, 2018). The

Analyte*	Unit	Method	Reagent
Total bilirubin and conjugated (TB and CB)	μM	DMSO. Colorimetric ¹	1001044
Glucose (GLU)	mM	Trinder. GOD-POD ²	1001190
Urea	mM	Urease-GLDH. Kinetic UV ³	1001333
Aspartate aminotransferase (AST)	U/L	NADH. Kinetic UV. IFCC rec. Liquid ⁴	MI41264
Albumin (ALB)	g/L	Colorimetric ⁵	1001020
Alanine aminotransferase (ALT)	U/L	NADH. Kinetic UV. IFCC rec. Liquid ⁶	41282
Cholesterol (CHOL)	mM	CHOD-POD. Liquid ⁷	41020
γ -glutamyl transferase (γ -GT)	U/L	Carboxy substrate. Kinetic. Liquid ⁸	41292

*Official abbreviation of the International Union of Pure and Applied Chemistry; ¹Bilirubin-azobilirubin by diazotized sulfanilic acid; ²Glucose oxidase (GOD)-Peroxidase (POD); ³Urease- α -Ketoglutarate-Glutamate dehydrogenase (GLDH); ⁴ α -ketoglutarate-glutamate and oxalacetate-malate dehydrogenase (MDH) and NADH; ⁵Bromocresol green; ⁶ α -ketoglutarate-glutamate and pyruvate-lactate dehydrogenase (LDH) and NADH; ⁷Cholesterol esterase (CHE)-Cholesterol oxidase (CHOD)-Peroxidase (POD); ⁸L- γ -Glutamyl-3-carboxy-4-nitroanilide-Glycylglycine; ⁸Spinreact., Girona, Spain.

The precision and reliability of the techniques were controlled using lyophilized control serum (SPINTROL NORMAL 1002100; Spinreact., Girona, Spain). Milk pH was measured with a (HI981032 pH meter; HANNA instruments., Mexico). The milk fat, SNF, milk density, milk protein and CRP were determined with ultrasonography (EKOMILK ULTRA Milkana KAM98-2A; milk analyzer., Bulgaria), which uses low power ultrasound (≥ 100 KHz) and low intensity ($\leq 1\text{W/cm}^2$). Ultrasound propagated through the milk as mechanical waves causing alternating compressions and decompressions (Soltani *et al.*, 2019). These waves interacted with the biochemical bonds present in milk fat and milk protein, stimulating their vibration and quantification (Bhargava *et al.*, 2021). The SCC was performed with infrared spectrophotometry (Ekomilk Scan; Ekomilk., Bulgaria). The latter dissolved the cell and nuclear membranes, allowing their contents to escape and forming a gel that increased the viscosity of the milk sample. The solution was then passed through the mixing capillary, and the flow time of the milk sample was recorded, which is proportional to the somatic cells/mL of milk. According to Piccinini *et al.* (2004) and Zecconi *et al.* (2019),

the infection risk scale will be categorized as follows:

- Up to 100×1000 cells/mL: healthy, normal milk
- From 100 to 200×1000 cells/mL: suspected, upper physiological level, and
- More than 200×1000 cells/mL: mastitis, abnormal milk.

The statistical procedure used to calculate the 95% confidence intervals for the different liver analytes was determined by (PROC TTEST; SAS, System, v. 8.2, Cary, NC). The recommendations of the International Federation of Clinical Chemistry were followed. The first step was the detection of outliers for each analyte and values more than three standard deviations from the mean were discarded. To measure the distribution and behavior of the different values within the sample, the resulting data set was analyzed using the Kolmogorov-Smirnov test. The percentiles: P_{10} - P_{90} and P_{25} - P_{75} were determined by PROC UNIVARIATE NORMAL PLOT (SAS, 2001). The comparison between groups (1st third of lactation vs. 2nd third of lactation) was assessed by PROC ANOVA (SAS, 2001), together with the Tukey and cldiff

Tukey post-hoc test. The between liver blood markers, parameters, and between liver with milk parameters are identified by PROC CORR (SAS, 2001). Regression Analysis PROC REG (SAS, 2001), whose statistical significance is presented by Pearson's correlation coefficient and a value of ($p < 0.05$), was used. The following model was tested:

$$Y = \beta_0 + \beta_1 \cdot X_1 + e_1$$

where:

Y = milk fat, SNF, milk density, milk protein, pH, and SCC,

$X_1, \dots, X_k$ = TB, CB, GLU, Urea, AST, ALB, ALT, CHOL, and γ -GT,

$\beta_1, \dots, \beta_k$ = partial regression coefficient for each X_1 ,

β_0 = intercept on the Y axis, and

e_1 = error of the estimate.

The analysis of the Outlier values was executed with a macro that uses the robust Mahalanobis distance, setting an importance to the data whose value has a probability of being (χ^2) $< p$. Regression diagnosis for main assumptions was. Kolmogorov-Smirnov test was used to check the normality of residuals. Homoscedasticity was checked by plotting residual versus predicted values, and the Durbin-Watson test was employed to check for error uncorrelation.

ETHICAL ASPECTS

All procedures used were approved by the Bioethics and Animal Welfare Committee of the Faculty of Veterinary Medicine and Animal Husbandry of the University of Colima (Assessment record: No. 002/2023).

RESULTS

Two observations with outliers were discarded. Descriptive statistics for the traits investigated are reported in Table 2.

Table 2. Descriptive statistics of single test-day, for different liver blood markers and milk parameters in intensive dairy farms, $n = 124$ samples

124 Samples																
	$\bar{x} \pm DE$			Reference ^a		CI ^b		P ₁₀ -P ₉₀		P ₂₅ -P ₇₅		KS ^c				
Liver blood markers																
TB, μM	6.63	$\pm$	0.70	0.17	-	8.55	6.51	-	6.76	5.71	-	7.56	6.15	-	7.11	>0.15
CB, μM	5.56	$\pm$	0.52	0.68	-	7.52	5.47	-	5.66	4.94	-	6.26	5.24	-	5.88	>0.15
GLU, mM	3.84	$\pm$	0.33	2.50	-	4.16	3.78	-	3.90	3.45	-	4.21	3.61	-	4.07	>0.15
Urea, mM	6.37	$\pm$	0.83	7.14	-	10.70	6.22	-	6.52	5.37	-	7.59	5.83	-	6.95	0.07
AST, U/L	83.34	$\pm$	21.33	78	-	132	79.55	-	87.14	56.51	-	112.1	71.67	-	97.81	0.10
ALB, g/L	33.01	$\pm$	2.39	30.30	-	35.50	32.58	-	33.43	29.87	-	36.00	31.37	-	34.68	>0.15
ALT, U/L	22.04	$\pm$	4.59	11	-	40	21.22	-	22.85	16.12	-	27.62	18.81	-	26.00	>0.15
CHOL, mM	3.52	$\pm$	1.03	2.07	-	3.50	3.34	-	3.71	2.16	-	4.85	2.96	-	4.13	0.08
γ -GT, U/L	20.97	$\pm$	3.65	6.10	-	19.40	20.32	-	21.62	16.25	-	25.88	18.37	-	23.87	>0.15
Milk parameters																
Milk fat, %	3.94	$\pm$	0.69	3.70	-	4.10	3.82	-	4.06	3.00	-	4.94	3.48	-	4.35	0.09
SNF, %	8.27	$\pm$	0.51	8.10	-	8.80	8.17	-	8.36	7.62	-	9.03	7.95	-	8.55	>0.15
Milk density, g/mL	1.01	$\pm$	0.01	1.01	-	1.04	1.00	-	1.01	0.99	-	1.03	1.00	-	1.02	>0.15
Milk protein, %	3.13	$\pm$	0.20	3.10	-	3.50	3.09	-	3.13	2.86	-	3.44	2.99	-	3.25	>0.15
SCC, x 1000/mL	100.55	$\pm$	9.83	100	-	120	98.80	-	102.3	86.40	-	114.6	92.98	-	107.4	0.06
pH	6.55	$\pm$	0.41	6.40	-	6.80	6.47	-	6.62	6.05	-	7.16	6.25	-	6.86	0.06
CRP, Horvert °H	-0.55	$\pm$	0.01	-0.50	-	-0.59	-0.55	-	-0.54	-0.57	-	-0.53	-0.56	-	-0.54	<0.01

TB: total bilirubin, CB: conjugated bilirubin, GLU: glucose, AST: aspartate aminotransferase, ALB: albumin, ALT: alanine aminotransferase, CHOL: cholesterol, γ -GT: γ -glutamyl-transferase, SNF: solids-non-fat, SCC: somatic cell count; pH: hydrogen potential; CRP: cryoscopic point; ^a(Kaneko *et al.*, 2008); ^b95% confidence interval; ^cKolmogorov-Smirnov normality test, $p \geq 0.05$.

The comparison between groups (1st third of lactation vs. 2nd third of lactation) showed

differences in GLU, AST, ALB, ALT, milk fat, SNF, milk protein and CRP (Table 3).

studentized range test, for different liver blood markers and milk parameters in intensive 62 Holstein cows/group

	Comparison by group	Difference between means	Confidence limits	Different group means
Liver blood markers				
TB, μM	1 - 2	0.04	-0.21 0.29	
	2 - 1	-0.04	-0.29 0.21	
CB, μM	1 - 2	0.00	-0.18 0.19	
	2 - 1	-0.00	-0.19 0.18	
GLU, mM	1 - 2	0.15	0.04 0.27	***
	2 - 1	-0.15	-0.27 -0.04	***
Urea, mM	1 - 2	0.02	-0.27 0.31	
	2 - 1	-0.02	-0.31 0.27	
AST, U/L	1 - 2	10.03	2.63 17.43	***
	2 - 1	-10.03	-17.43 -2.63	***
ALB, g/L	1 - 2	2.43	1.69 3.17	***
	2 - 1	-2.43	-3.17 -1.69	***
ALT, U/L	1 - 2	2.16	0.57 3.76	***
	2 - 1	-2.16	-3.76 -0.57	***
CHOL, mM	1 - 2	0.11	-0.25 0.48	
	2 - 1	-0.11	-0.48 0.25	
γ -GT, U/L	1 - 2	0.26	-1.03 1.56	
	2 - 1	-0.26	-1.56 1.03	
Milk parameters				
Milk fat, %	1 - 2	0.33	0.09 0.56	***
	2 - 1	-0.33	-0.56 -0.09	***
SNF, %	1 - 2	0.45	0.29 0.62	***
	2 - 1	-0.45	-0.62 -0.29	***
Milk density, g/mL	1 - 2	0.00	-0.00 0.00	
	2 - 1	-0.00	-0.00 0.00	
Milk protein, %	1 - 2	0.22	0.16 0.28	***
	2 - 1	-0.22	-0.28 -0.16	***
SCC, x 1000/mL	1 - 2	1.44	-2.05 4.94	
	2 - 1	-1.44	-4.94 2.05	
pH	1 - 2	0.12	-0.02 0.26	
	2 - 1	-0.12	-0.26 0.02	
CRP, Horvert °H	1 - 2	0.00	0.00 0.01	***
	2 - 1	-0.00	-0.01 -0.00	***

TB: total bilirubin, CB: conjugated bilirubin, GLU: glucose, AST: aspartate aminotransferase, ALB: albumin, ALT: alanine aminotransferase, CHOL: cholesterol, γ -GT: γ -glutamyl-transferase, SNF: solids-non-fat, SCC: somatic cell count; pH: hydrogen potential; CRP: cryoscopic point; ¹1st third of lactation [DIM: 42 $\pm$ 15 d p.p.; Milk Production (Mean $\pm$ SD): 35.36 $\pm$ 1.49 kg·d⁻¹], ²2nd third of lactation [DIM: 91 $\pm$ 17 d p.p.; Milk Production (Mean $\pm$ SD): 16.34 $\pm$ 1.23 kg·d⁻¹]; Significant comparisons at the 0.05 level are indicated by ***

The interaction between liver blood markers is reported in Table 4. Positive relationships ($p < 0.05$) between TB with CB and CHOL were

calculated. A negative relationship ($p < 0.05$) between ALT and γ -GT was identified.

correlation coefficient between liver blood markers, $n = 124$ Holstein cows

	GLU	Urea	AST	ALB	ALT	CHOL	γ -GT
*	0.27	0.38	0.17	0.09	0.10	0.94*	-0.06
	0.46	0.60	0.25	-0.01	-0.03	0.66	0.04
	GLU	0.19	0.31	-0.47	-0.63	0.13	0.48
		Urea	0.17	-0.01	-0.01	0.24	0.00
			AST	-0.38	-0.16	0.13	-0.04
				ALB	0.61	0.11	-0.11
					ALT	0.15	-0.85*
						CHOL	-0.12
							γ -GT

TB: total bilirubin, CB: conjugated bilirubin, GLU: glucose, AST: aspartate aminotransferase, ALB: albumin, ALT: alanine aminotransferase, CHOL: cholesterol, γ -GT: γ -glutamyl-transferase; * $p < 0.05$.

According to the Kolmogorov-Smirnov normality test the CRP is not normally distributed (Table 2). Therefore, it was excluded from the relationship models. The interaction between milk parameters is reported in Table 5.

A positive relationship ($p < 0.05$) between SNF and milk protein was identified. Negative relationships ($p < 0.05$) between milk fat with SNF, milk protein, and SCC were quantified.

Table 5. Pearson correlation coefficient between milk parameters, $n = 124$ Holstein cows

	SNF	Milk density	Milk protein	pH	SCC
Milk fat	-0.81*	0.05	-0.76*	-0.06	-0.82*
	SNF	-0.12	0.98*	0.11	0.71*
		Milk density	-0.12	-0.08	-0.02
			Milk protein	0.13	0.65
				pH	0.04
					SCC

SNF: non-fat solids, pH: hydrogen potential, SCC: somatic cell count; * $p < 0.05$.

The interaction between liver blood markers and milk parameters is reported in Table 6. Positive relationships ($p < 0.05$) between milk fat and GLU; SNF and ALB; milk protein and ALB; pH

and urea; and SCC and ALT were calculated. A negative relationship ($p < 0.05$) between milk fat and ALT was identified.

Table 6. Pearson correlation coefficient between liver blood markers and milk parameters, $n = 124$ Holstein cows

	TB	CB	GLU	Urea	AST	ALB	ALT	CHOL	γ -GT
Milk fat	0.19	0.32	0.78*	0.12	0.36	-0.62	-0.82*	0.11	0.62
SNF	-0.12	-0.24	-0.66	-0.12	-0.40	0.76*	0.67	-0.06	-0.34
Milk density	0.04	-0.01	0.01	-0.05	0.06	0.08	-0.02	0.08	-0.09
Milk protein	-0.11	-0.22	-0.63	-0.11	-0.41	0.77*	0.64	-0.06	-0.29
pH	0.35	0.55	0.04	0.97*	0.07	0.17	0.14	0.22	-0.06
SCC	-0.20	-0.27	-0.65	-0.11	-0.27	0.52	0.71*	-0.16	-0.54

TB: total bilirubin, CB: conjugated bilirubin, GLU: glucose, AST: aspartate aminotransferase, ALB: albumin, ALT: alanine aminotransferase, CHOL: cholesterol, γ -GT: γ -glutamyl-transferase, SNF: solids-non-fat, pH: hydrogen potential, SCC: somatic cell count; * $p < 0.05$.

DISCUSSION

tions of blood parameters e present study (Table 2) were within the reference ranges reported for dairy cows without clinical signs of disease (Kaneko *et al.*, 2008). As expected, there were differences in milk parameters between 1st third of lactation and 2nd third of lactation groups. SNF and milk protein were higher in 1st third of the lactation. However, milk fat was lower in 1st third of lactation (Table 3). The GLU concentration was not significantly correlated with other investigated liver blood markers. This may be due to the high compensation of the gluconeogenesis process by the regulation of glucagon and insulin (Larsen and Kristensen, 2013). In dairy cows, hepatic gluconeogenesis fulfills more than 80% of the GLU requirement (Aschenbach *et al.*, 2010). The main substrate for gluconeogenesis in the liver is propionate, which provides more than 60% of the carbon source (Larsen and Kristensen, 2013). When the demand for GLU increases, ruminant liver can

also utilize amino acids, lactate, and glycerol for gluconeogenesis, providing 11 to 16% of the carbon source (Andjelic *et al.*, 2022). In our study, the GLU level was lower in the 1st third of lactation compared to the 2nd third of lactation. The high-yielding cows require large amounts of GLU as a substrate and as a source of energy for lactose synthesis (Larsen and Kristensen, 2013). Therefore, the GLU levels observed in 1st third of lactation may be associated with an increased lactose synthesis (Azzout *et al.*, 2007). About 80% of lactose is produced from GLU, and the remaining 20% is produced from acetate (Wang *et al.*, 2024). Also, lactose is the major osmoregulator of mammary water uptake and, consequently, determines milk volume (Aschenbach *et al.*, 2010). Also, GLU was positively related to milk fat (Table 6, Fig. 1). This subordinate behavior can be explained by the fact that milk fat is composed of glycerol and fatty acids (Andjelic *et al.*, 2022). Glycerol is formed during glycolysis, while fatty acids are formed during ruminal fermentation of plant fibers (Aschenbach *et al.*, 2010).

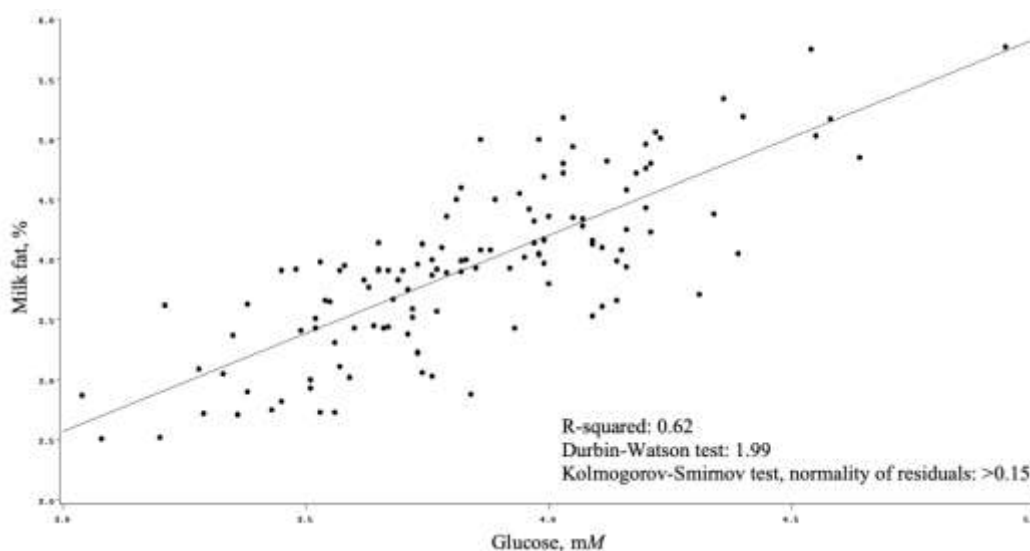


Figure 1. Relationship between glucose and milk fat, $n = 124$ Holstein cows. Milk fat (•); predicted response (—).

Hemoglobin is a protein that is present in red blood cells (Guerra *et al.*, 2021). In its heme group, transported oxygen, to interchange in the lungs with carbon dioxide (Andjelic *et al.*, 2022). This interchange caused the rupture of the heme group's bonds and its transformation into biliverdin, which is converted into unconjugated or indirect bilirubin (UCB) (Katica *et al.*, 2024).

The UCB is water-insoluble and enters circulation bound to ALB (Andjelic *et al.*, 2022). In the hepatocyte, glucuronic acid is added to UCB (conjugation) to render it water-soluble (CB) (Guerra *et al.*, 2021). The TB is the sum of CB and UCB (Kaneko *et al.*, 2008). This dependence would explain the quantified relationship between TB and CB (Fig. 2).

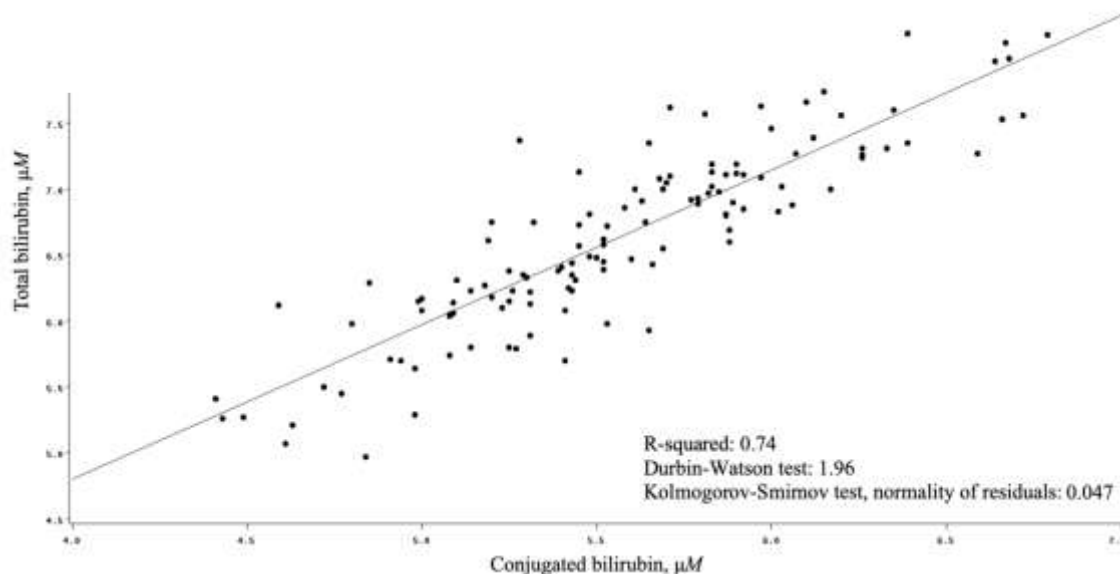


Figure 2. Relationship between conjugated bilirubin and total bilirubin, $n = 124$ Holstein cows. Total bilirubin ($\bullet$); predicted response (—).

The CB is excreted into bile or recirculated back to the bloodstream, where it is filtrated by the kidneys and excreted through urine (Katica *et al.*, 2024). All liver lesions induce a decrease in the hepatocyte cell count, which may cause hyperbilirubinemia (Bruckmaier and Gross, 2017), a condition not observed in this study (Table 2). BT and CB are well-established blood markers that are routinely included in tests for cows with liver dysfunction (García *et al.*, 2020). However, BT and CB are not a specific marker of liver function, so quantification of other blood markers is also necessary, e.g. AST, ALT, and γ -GT (Pinedo and Melendez, 2022). In comparison with other stages of lactation, during the 1st third, cows experience a period of high energy requirements, associated with insufficient feed intake (Pinedo and Melendez, 2022). To compensate for this nutrient deficiency, the cow mobilizes its body's fat and protein reserves (Haussler *et al.*, 2022). This situation causes a Negative Energy Balance (NEB) (Han *et al.*, 2012), which alters the enzymatic activity of AST, ALT, and γ -GT (Table 3). When the liver is diseased or damaged, additional AST and ALT are released into the bloodstream (Bruckmaier and Gross, 2017). Therefore, the amount of AST and ALT in the blood is directly related to the extent of the tissue damage (Han *et al.*, 2012). However, the enzyme values of AST and ALT recorded in our study (Table 2) did not exceed

the international reports for healthy cows. Therefore, the liver functions of cows were not disturbed (Han *et al.*, 2012). It may be that hepatic gluconeogenesis is minimizing lipid metabolism disorders in BEN, such as fatty liver and ketosis (Aschenbach *et al.*, 2010; Andjelic *et al.*, 2022). In contrast to AST, ALT is a cytoplasmic enzyme and in cattle, is involved in the metabolism of proteins, catalyzing a reversible reaction of α -alanine synthesis (Han *et al.*, 2012). The γ -GT is an enzyme that transfers a γ -glutamyl group present in glutamic acid, a key amino acid in the cellular metabolism of proteins (Paiano *et al.*, 2020). An elevated serum γ -GT appears to be a sensitive specific indicator of liver damage, impaired biliary secretion, intense protein translation, or very high gluconeogenesis, making it a useful diagnostic aid (Wang *et al.*, 2019; McFadden, 2020; Zhang *et al.*, 2020). The enzyme values of γ -GT recorded in our study (Table 2) did not exceed the international reports for healthy cows. Therefore, the liver functions of cows were not disturbed. It should also be noted that γ -GT shows greater specificity in cattle liver than AST and ALT (McFadden, 2020).

The recorded activity of the ALT parameter was negatively correlated with the activity of γ -GT (Table 4). Such a negative correlation can often be observed in cows with high milk production,

g the increase in mitochondrial *al.*, 2012). The ALB is positively protein and SNF (Figures 3 and nation is because ALB serves as a carrier for molecules with low water content, including SNF, fat-soluble hormones, and free

fatty acids. Also, the serum ALB concentration can be expressed in terms of milk protein content because it is a common component of body fluids, both in the plasma and milk of cows (Lieske *et al.*, 2005).

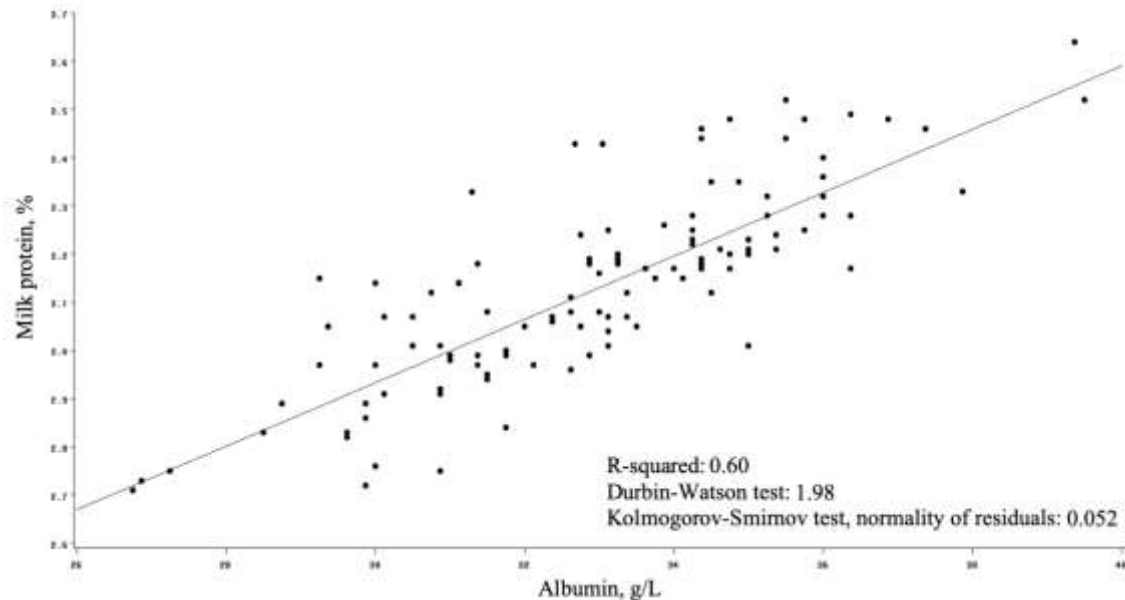


Figure 3. Relationship between albumin and milk protein, $n = 124$ Holstein cows. Albumin (•); predicted response (—).

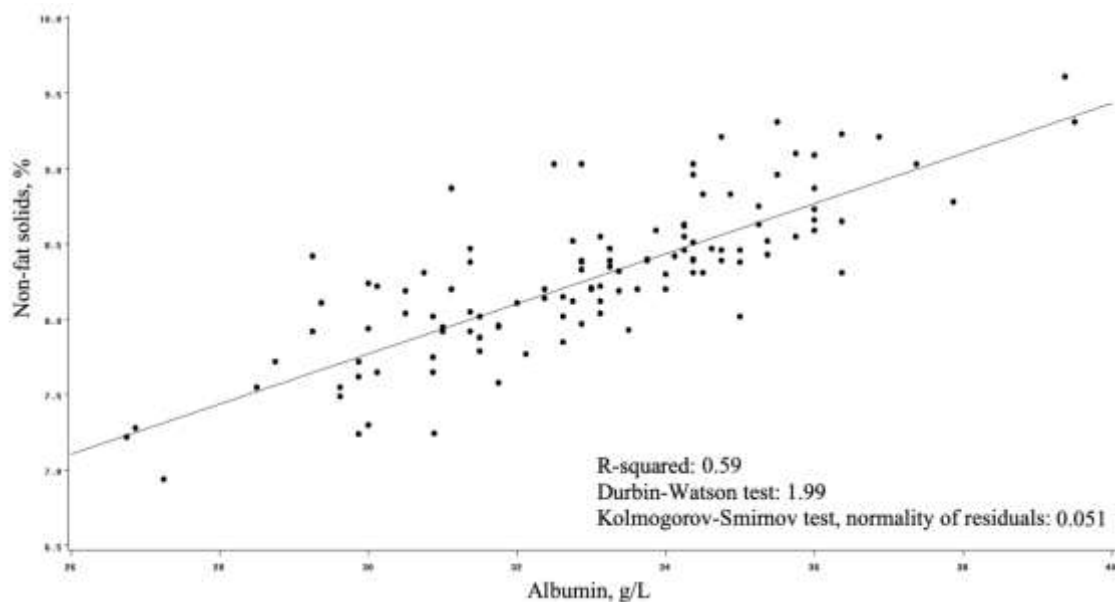


Figure 4. Relationship between albumin and non-fat solids, $n = 124$ Holstein cows. Albumin (•); predicted response (—).

the ammonia reaching the liver has exogenous, absorbed through the rumen and endogenous, originating from the animal organism (Jin *et al.*, 2018). Ammonia is absorbed by the ruminal wall and reaches the periportal hepatocytes, which contain ureagenic enzymes for Urea synthesis. In mammals, Urea represents the molecule used for excretion of ammonia (Jin *et al.*, 2018). Its

concentration in blood is highly affected by dietary factors, especially the amount of protein ingested and the protein/energy ratio (Weiner *et al.*, 2015). The Urea is positively related to pH (Figure 5). This subordination is due to the hydrolysis of Urea, catalyzed by the enzyme urease, which produces ammonia, which reacts with water to form ammonium hydroxide, thus raising the pH (Weiner *et al.*, 2015).

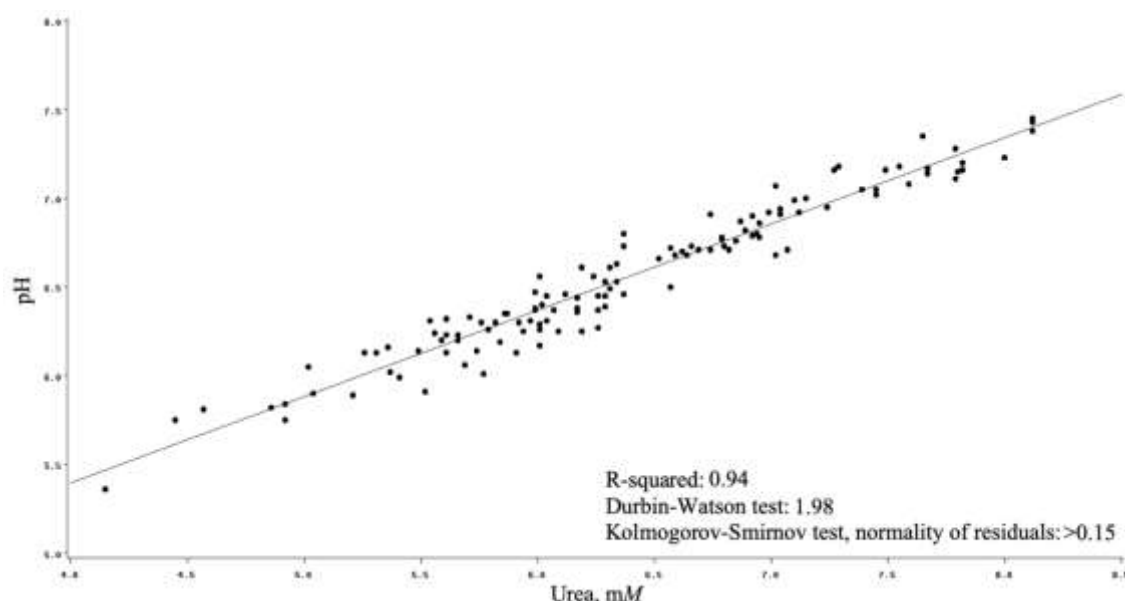


Figure 5. Relationship between urea and pH, $n = 124$ Holstein cows. Urea ($\bullet$); predicted response (—).

The negative relationship between milk fat and SNF (Table 5), is obvious since SNF does not include fats. The positive correlation ($p < 0.05$) between milk protein and SNF recorded in our study is expected, as milk proteins make a significant contribution to this mass (Macrae *et al.*, 2019). Finally, the negative relationship between SCC and milk fat (Table 5) is consistent with the results of other authors (Pedraza *et al.*, 2000). Clinical and subclinical mastitis increase the milk SCC (Summer *et al.*, 2015). This marker is routinely used at the herd level as an indicator of udder health (Bisutti *et al.*, 2022). The SCC includes leukocytes and epithelial cells (Summer *et al.*, 2015). The leucocyte populations reflect the intramammary infection severity (Bobbo *et al.*, 2016). In healthy udders, lymphocytes and macrophages prevail (Pegolo *et al.*, 2021). However, the fairly low SCC in milk (Table 2), indicated quite good mammary gland health status (Summer *et al.*, 2015).

CONCLUSIONS

The present study evaluated the relationships between liver blood markers and milk parameters in cows during the 1st and 2nd third of lactation. Glucose, aspartate aminotransferase, albumin, alanine aminotransferase, milk fat, solids-non-fat, milk protein and cryoscopic point showed differences between groups. Total bilirubin, conjugated bilirubin, urea, cholesterol, γ -glutamyl-transferase, milk density, somatic cell count, and pH do not show differences between the 1st and 2nd third of lactation. Positive relationships between total bilirubin with conjugated bilirubin and cholesterol, solids-non-fat and milk protein, milk fat and glucose, solids-non-fat and albumin, milk protein and albumin, pH and urea, and somatic cell count with alanine aminotransferase were found. The negative relationship between alanine aminotransferase and γ -glutamyl-transferase, milk fat with non-fat solids, milk protein, and somatic cell count, and

alanine aminotransferase were use the liver has a variety of eral liver function tests are used tics and management of liver diseases. Determination of liver blood markers and milk parameters can provide valuable information regarding dairy cow's nutrition and physiological status.

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CONTRIBUTORS

R.O.F. Prado. Writing original draft and correction. A.C.O. Aguilar. Writing original draft and supervision. R.J.A. Hernández. Data curation and supervision. D.H. Parra. Review and correction. L.L. Galicia. Data curation and supervision. C.A.C. García. Study conception, writing original draft, review and correction.

DATA AVAILABILITY STATEMENT

Data-in-article – the research data are available within the article itself.

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